

Feature Name	Permutation Random Forest	Permutation CART	Permutation KNN	Decision Tree	Random Forest	XGBoost	Permutation XGBoost
age	0.010	0.111	0.171	0.080	0.089	0.041	0.044
sex	0.010	0.047	0.000	0.037	0.036	0.074	0.047
cp	0.050	0.152	0.000	0.270	0.134	0.253	0.093
trestbps	0.003	0.050	0.261	0.052	0.074	0.028	0.006
chol	0.003	0.071	0.358	0.088	0.079	0.034	0.039
fbs	0.000	0.007	0.000	0.010	0.009	0.026	0.000
restecg	0.000	0.015	0.000	0.019	0.020	0.030	0.006
thalach	0.008	0.053	0.357	0.043	0.120	0.031	0.020
exang	0.012	0.029	0.000	0.030	0.060	0.079	0.014
oldpeak	0.017	0.067	0.000	0.101	0.116	0.051	0.036
slope	0.000	0.024	0.000	0.034	0.049	0.067	0.001
ca	0.051	0.156	0.000	0.155	0.117	0.114	0.104
thal	0.042	0.093	0.000	0.081	0.097	0.172	0.060

Table 5. Sophisticated Results of Feature Importance with tree-based classifiers.

Rank	Feature	Borda Count
1	cp	83
2	ca	75
3	thal	63
4	age	59
5	chol	57
6	oldpeak	57
7	thalach	53
8	sex	49
9	exang	41
10	trestbps	39
11	slope	25
12	restecg	21
13	fbs	15

Table 6. Sophisticated Borda Count ranking.

Number of Features	Random Forest	Gradient Boosting	XGBoost	CatBoost	LightGBM
1	0.7600	0.7600	0.7600	0.7600	0.7600
2	0.7541	0.7522	0.7541	0.7512	0.7493
3	0.8468	0.8478	0.8459	0.8459	0.8488
4	0.9376	0.8937	0.9385	0.8780	0.9366
5	1.0000	0.9415	1.0000	0.8985	1.0000
6	1.0000	0.9639	1.0000	0.9259	1.0000
7	1.0000	0.9590	1.0000	0.9346	1.0000
8	0.9961	0.9610	1.0000	0.9337	1.0000
9	0.9961	0.9688	1.0000	0.9395	1.0000
10	1.0000	0.9746	1.0000	0.9415	1.0000
11	0.9961	0.9727	1.0000	0.9473	1.0000
12	1.0000	0.9746	1.0000	0.9571	1.0000
13	0.9961	0.9717	1.0000	0.9551	1.0000
Average	0.9248	0.9186	0.9460	0.8976	0.9457

Table 7. Sophisticated ML classification results with average accuracy across different feature subsets.